# WORKSHEET - General 2 Mathematics

**Topic Areas:**

**Algebra and Modelling**

AM5 – Non-linear relationships

- Exponential/Quadratic (Projectile)
- Inverse
- Perimeter/Area problem

**Teacher:** PETER HARGRAVES

**Source:** HSC exam questions

**Exam Equivalent Time:** 81 minutes

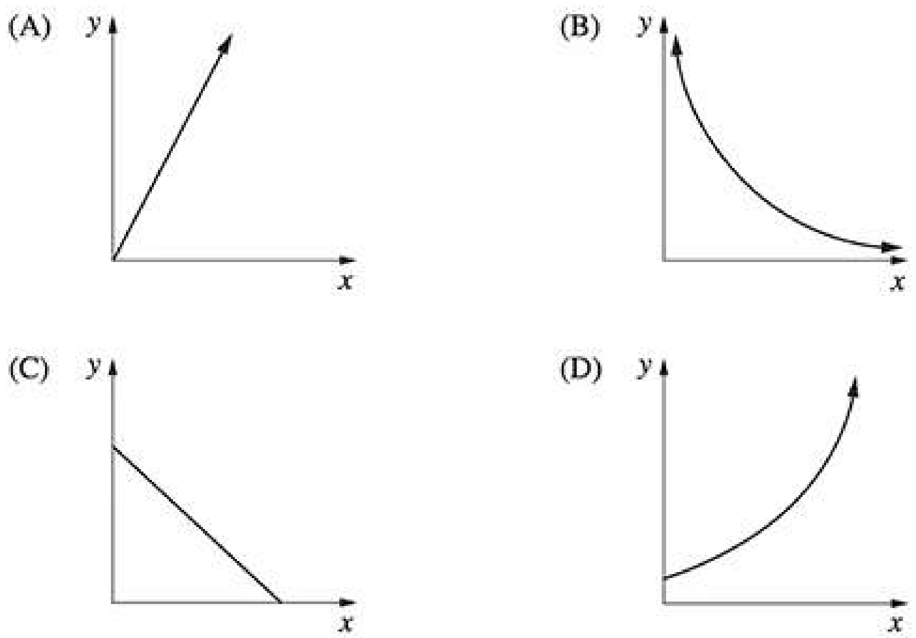
**Worked Solutions:** Included

*Note: Each question has designated marks. Use this information as both a guide to the question's difficulty and as a timing indicator, whereby each mark should equate to 1.5 minutes of working (examination) time.*

## Questions

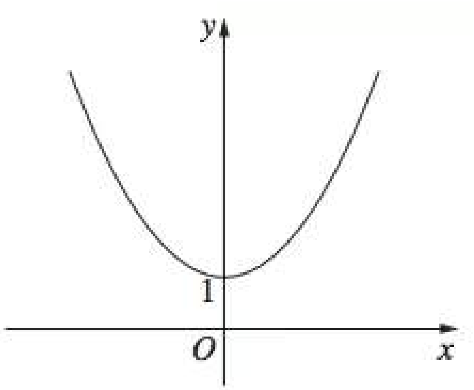
### 1. Algebra, 2UG 2008 HSC 4 MC

Which graph best represents $y = 3^x$?



## 2. Algebra, 2UG 2014 HSC 3 MC

The diagram shows the graph of an equation.




Which of the following equations does the graph best represent?

**(A)** $y = \dfrac{3}{x} + 1$

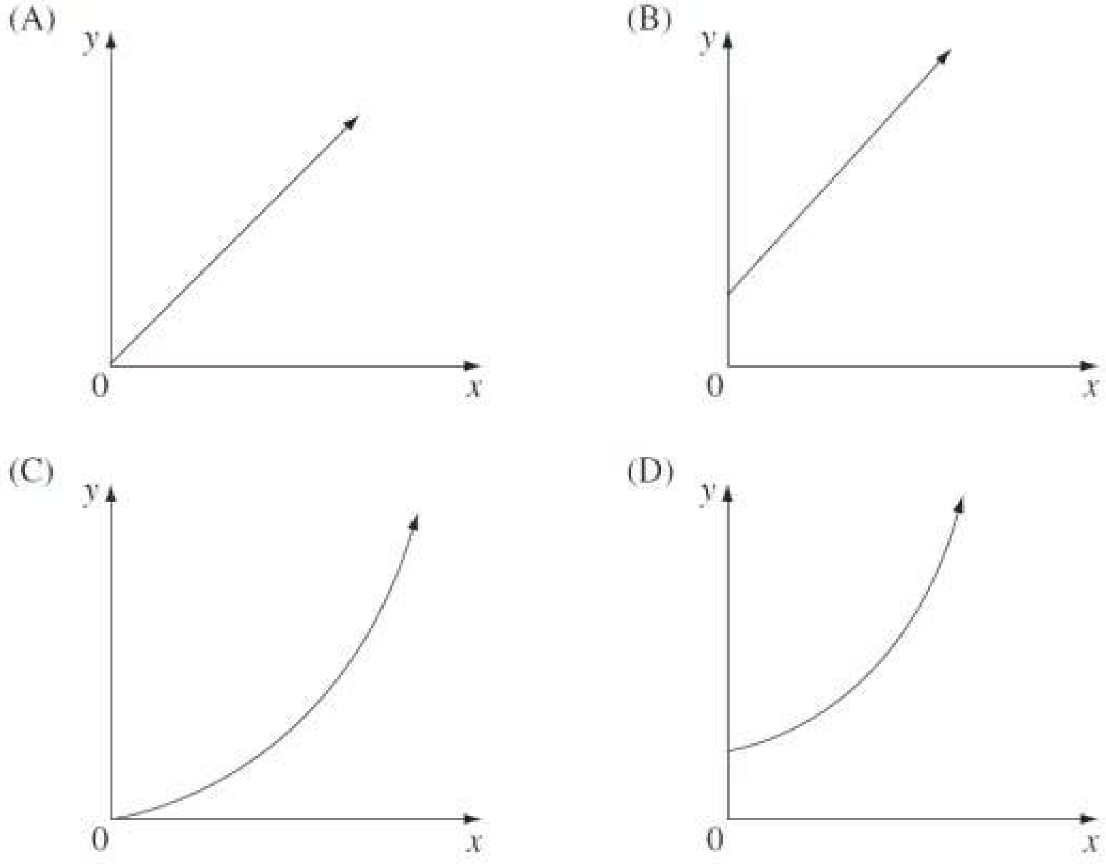
**(B)** $y = 3^x + 1$

**(C)** $y = 3x^2 + 1$

**(D)** $y = 3x^3 + 1$

## 3. Algebra, 2UG 2011 HSC 6 MC

Which of the following graphs best represents the equation $y = a^x$, where $a$ is a positive number greater than 1?

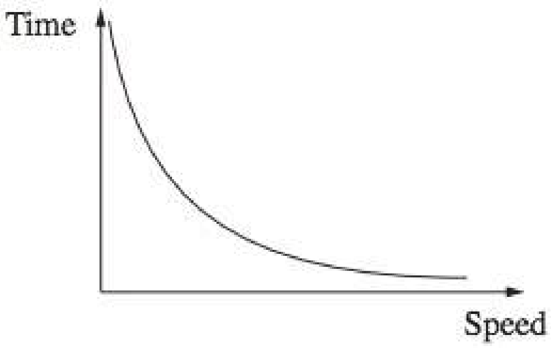


## 4. Algebra, 2UG 2009 HSC 16 MC

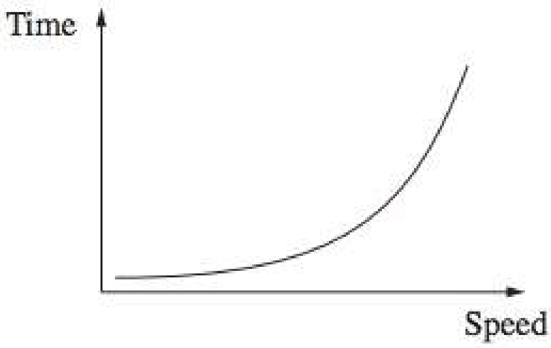
The time for a car to travel a certain distance varies inversely with its speed.

Which of the following graphs shows this relationship?

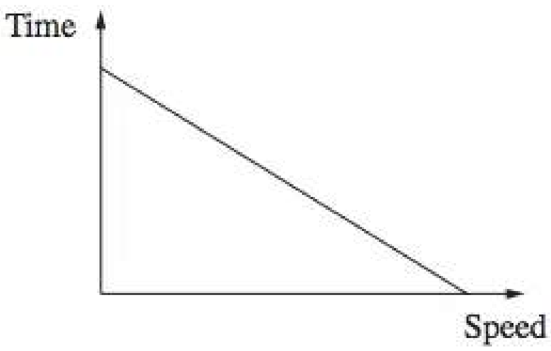
(A)



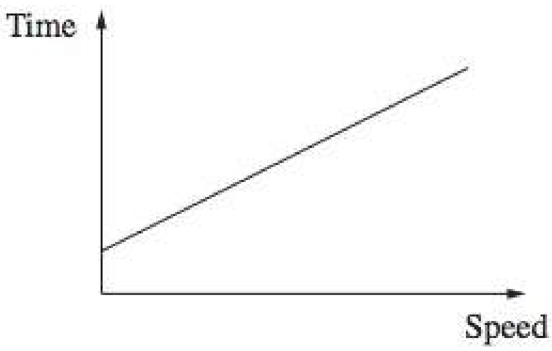

(B)




(C)




(D)




## 5. Algebra, 2UG 2010 HSC 13 MC

The number of hours that it takes for a block of ice to melt varies inversely with the temperature. At $30^\circ\mathrm{C}$ it takes $8$ hours for a block of ice to melt.

How long will it take the same size block of ice to melt at $12^\circ\mathrm{C}$?

**(A)** $3.2$ hours

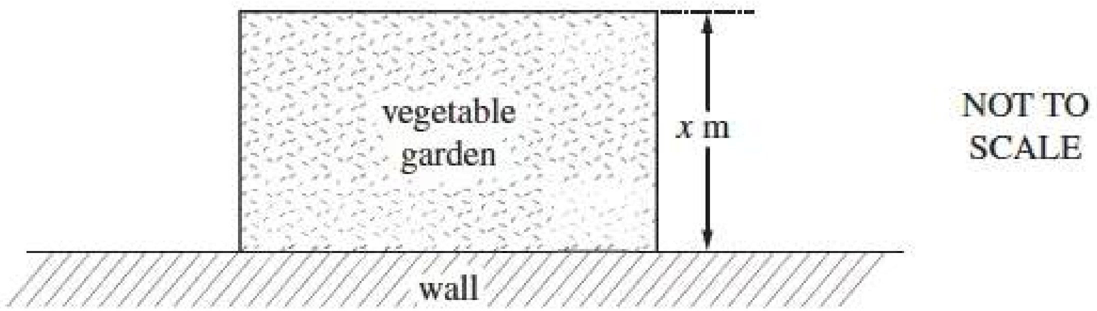
**(B)** $20$ hours

**(C)** $26$ hours

**(D)** $45$ hours

## 6. Algebra, 2UG 2013 HSC 22 MC

Leanne wants to build a rectangular vegetable garden in her backyard. She has $20$ metres of fencing and will use a wall as one side of the garden. The plan for her garden is shown, where $x$ metres is the width of her garden.




Which equation gives the area, $A$, of the vegetable garden?

**(A)** $A = 10x - x^2$

**(B)** $A = 10x - 2x^2$

**(C)** $A = 20x - x^2$

**(D)** $A = 20x - 2x^2$

## 7. Algebra, 2UG 2008 HSC 19 MC

The height of a particular termite mound is directly proportional to the square root of the number of termites.

The height of this mound is $35$ cm when the number of termites is $2000$.

What is the height of this mound, in centimetres, when there are $10\ 000$ termites?

**(A)** $16$

**(B)** $78$

**(C)** $175$

**(D)** $875$

## 8. Algebra, 2UG 2009 HSC 28c

The height above the ground, in metres, of a person's eyes varies directly with the square of the distance, in kilometres, that the person can see to the horizon.

A person whose eyes are $1.6 \text{ m}$ above the ground can see $4.5 \text{ km}$ out to sea.

How high above the ground, in metres, would a person's eyes need to be to see an island that is $15 \text{ km}$ out to sea? Give your answer correct to one decimal place. ***(3 marks)***

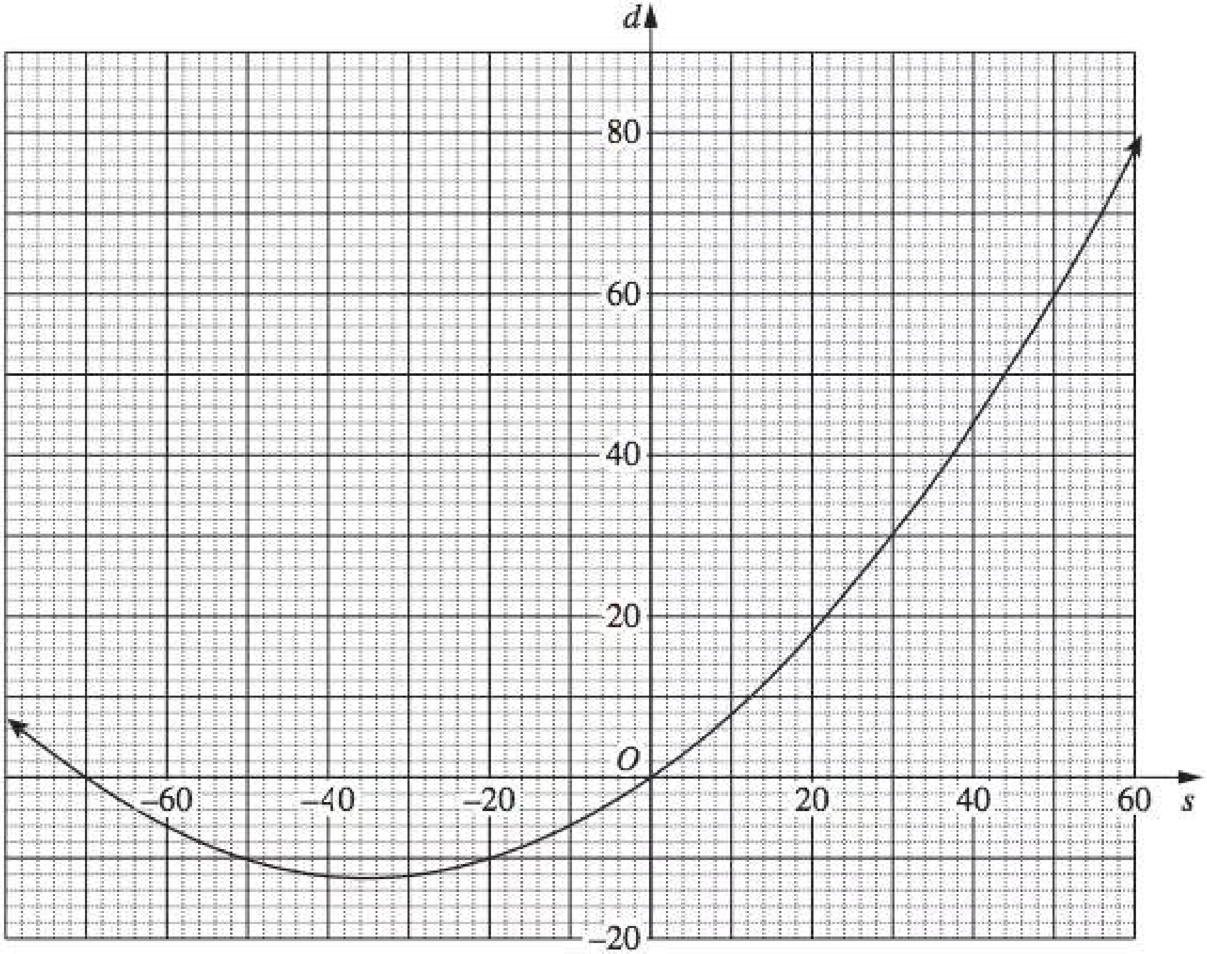
## 9. Algebra, 2UG 2009 HSC 28a

Anjali is investigating stopping distances for a car travelling at different speeds. To model this she uses the equation

$$d = 0.01s^2 + 0.7s,$$

where $d$ is the stopping distance in metres and $s$ is the car's speed in km/h.

The graph of this equation is drawn below.




**(i)** Anjali knows that only part of this curve applies to her model for stopping distances.

In your writing booklet, using a set of axes, sketch the part of this curve that applies for stopping distances. ***(1 mark)***

**(ii)** What is the difference between the stopping distances in a school zone when travelling at a speed of $40 \text{ km/h}$ and when travelling at a speed of $70 \text{ km/h}$? ***(2 marks)***

## 10. Algebra, 2UG 2011 HSC 28a

The air pressure, $P$, in a bubble varies inversely with the volume, $V$, of the bubble.

**(i)** Write an equation relating $P$, $V$ and $a$, where $a$ is a constant. ***(1 mark)***

**(ii)** It is known that $P = 3$ when $V = 2$.

By finding the value of the constant, $a$, find the value of $P$ when $V = 4$. ***(2 marks)***

**(iii)** Sketch a graph to show how $P$ varies for different values of $V$.

Use the horizontal axis to represent volume and the vertical axis to represent air pressure. ***(2 marks)***

## 11. Algebra, 2UG 2012 HSC 30c

In 2010, the city of Thagoras modelled the predicted population of the city using the equation

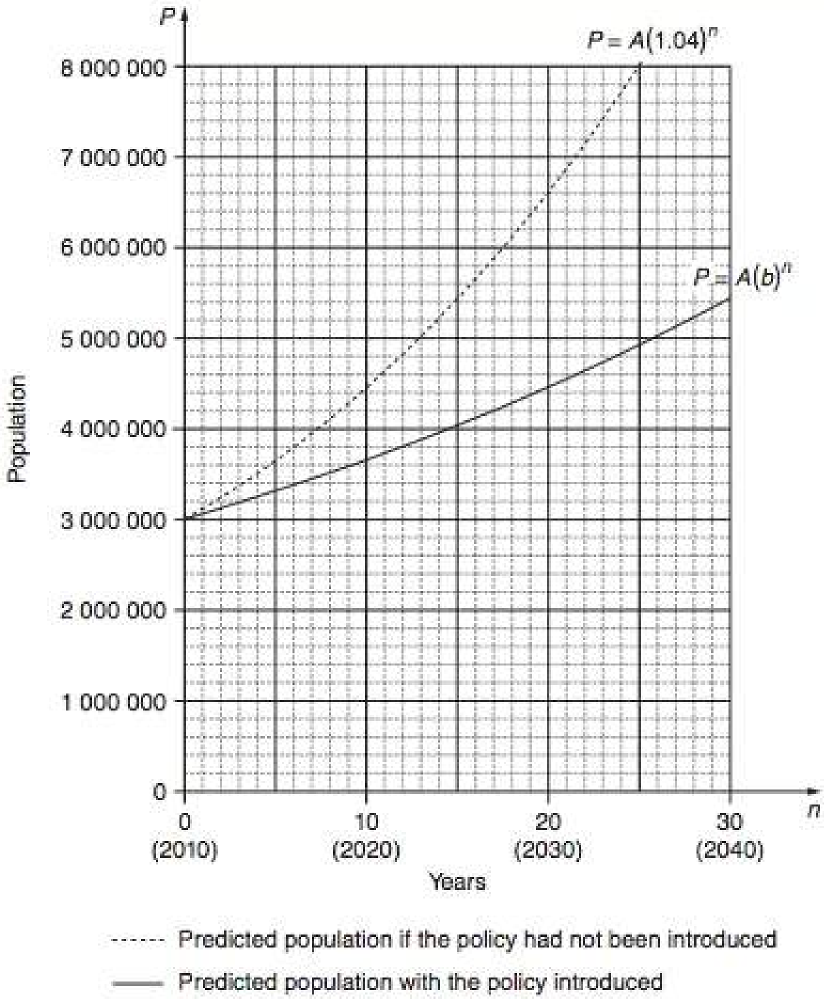
$$P = A(1.04)^n.$$

That year, the city introduced a policy to slow its population growth. The new predicted population was modelled using the equation

$$P = A(b)^n.$$

In both equations, $P$ is the predicted population and $n$ is the number of years after 2010.

The graph shows the two predicted populations.




**(i)** Use the graph to find the predicted population of Thagoras in 2030 if the population policy had NOT been introduced. ***(1 mark)***

**(ii)** In each of the two equations given, the value of $A$ is $3\ 000\ 000$.

What does A represent? ***(1 mark)***

**NB. Parts (iii) - (iv):** No longer in syllabus.

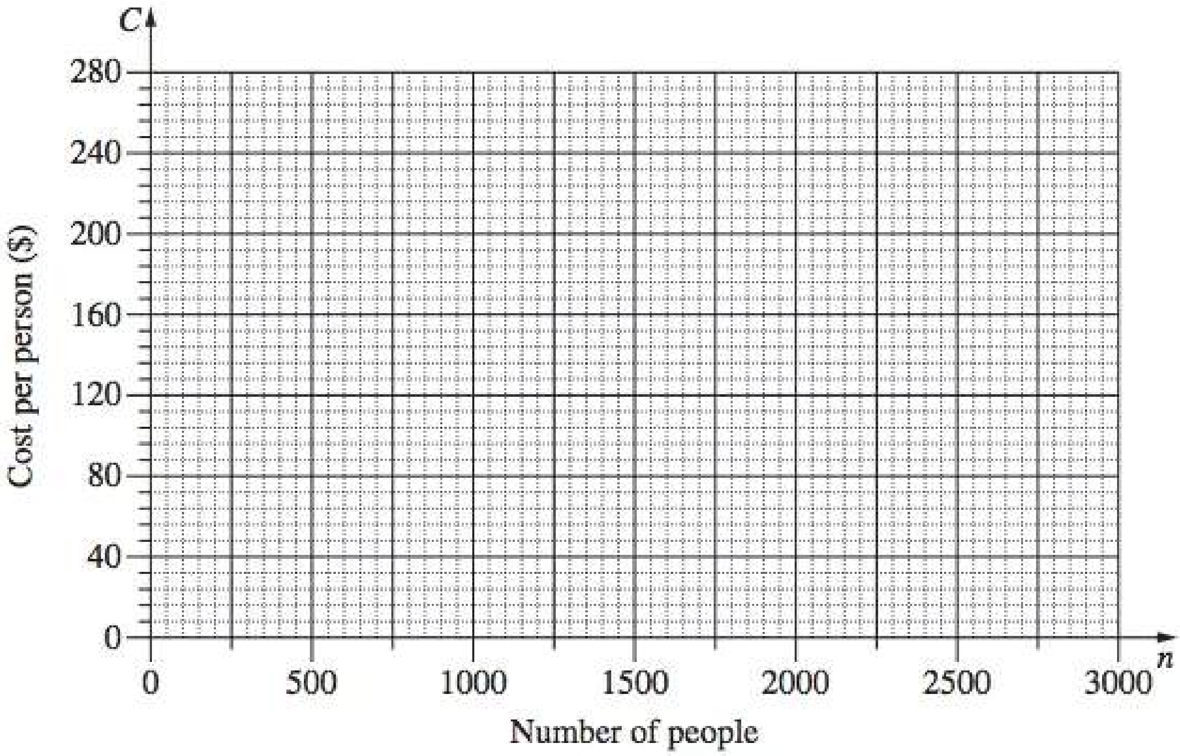
## 12. Algebra, 2UG 2014 HSC 29a

The cost of hiring an open space for a music festival is $\$120\,000$. The cost will be shared equally by the people attending the festival, so that $C$ (in dollars) is the cost per person when $n$ people attend the festival.

**(i)** Complete the table below by filling in the THREE missing values. ***(1 mark)***

| Number of people ($n$) | 500 | 1000 | 1500 | 2000 | 2500 | 3000 |
|---|---|---|---|---|---|---|
| Cost per person ($C$) | | | | 60 | 48 | 40 |

**(ii)** Using the values from the table, draw the graph showing the relationship between $n$ and $C$. ***(2 marks)***



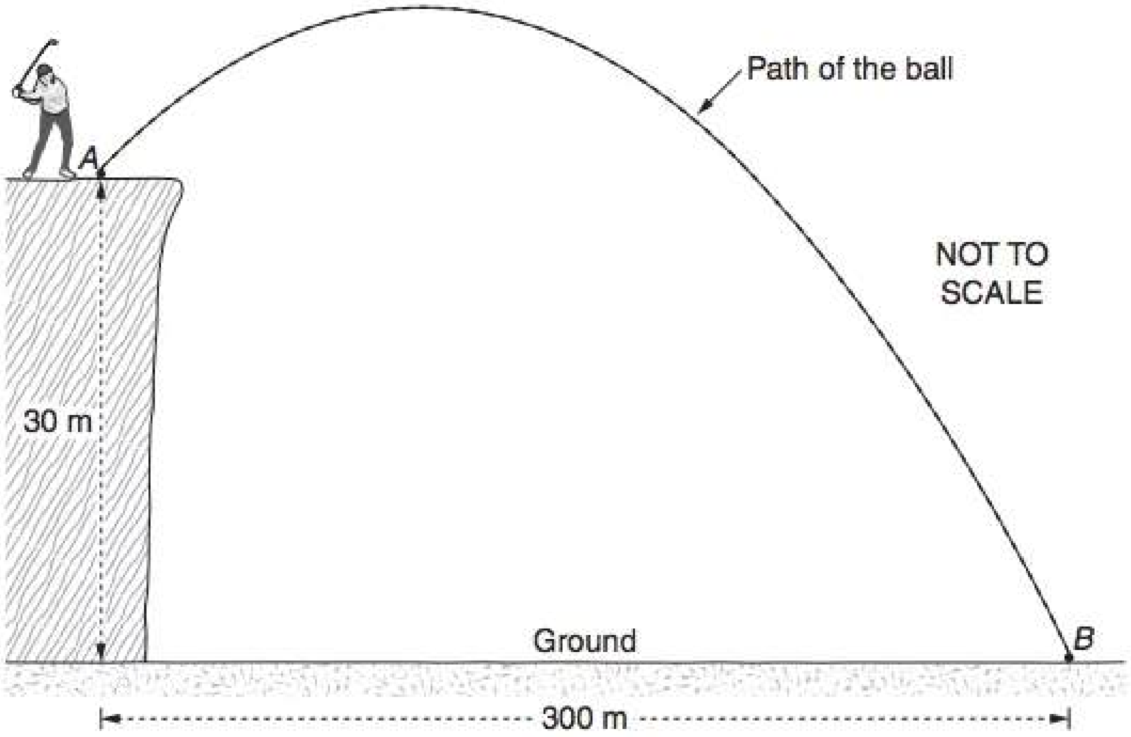

**(iii)** What equation represents the relationship between $n$ and $C$? ***(1 mark)***

**(iv)** Give ONE limitation of this equation in relation to this context. ***(1 mark)***

**(v)** Is it possible for the cost per person to be $\$94$? Support your answer with appropriate calculations. ***(1 mark)***

## 13. Algebra, 2UG 2012 HSC 30b

A golf ball is hit from point $A$ to point $B$, which is on the ground as shown. Point $A$ is $30$ metres above the ground and the horizontal distance from point $A$ to point $B$ is $300$ m.




The path of the golf ball is modelled using the equation

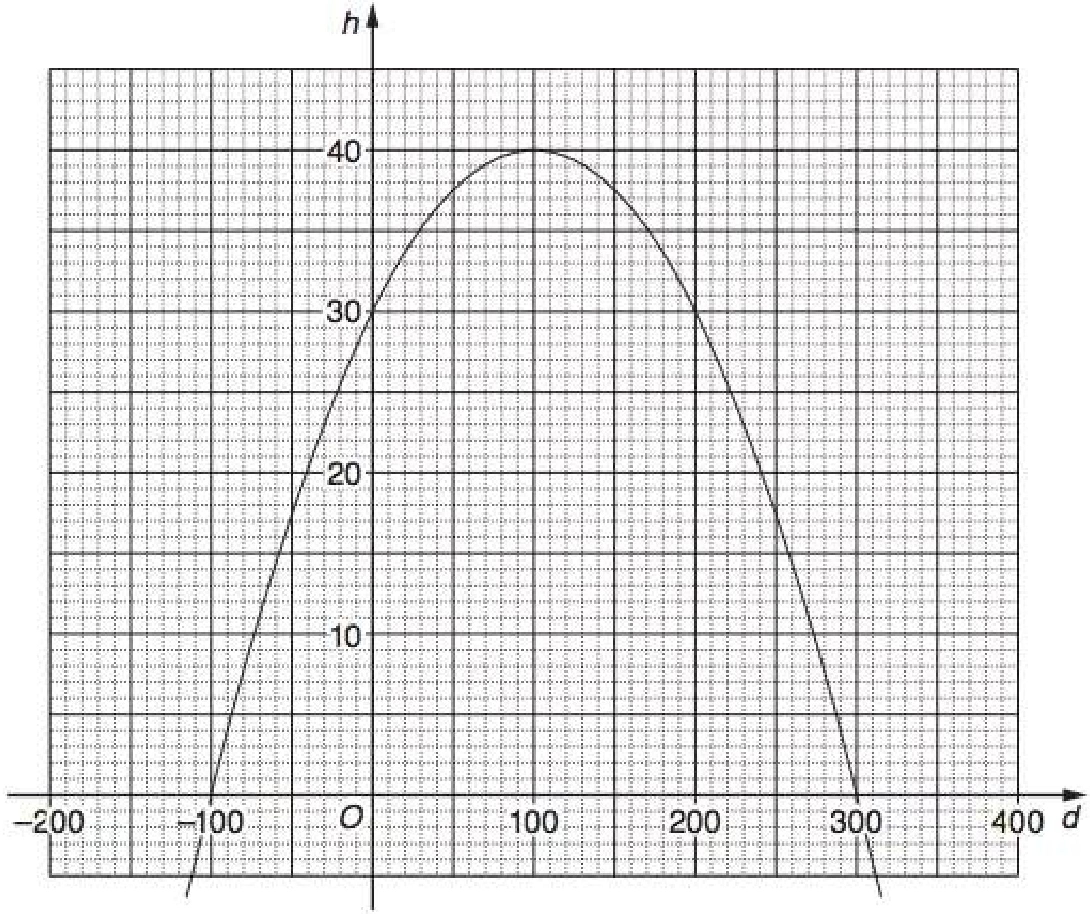
$$h = 30 + 0.2d - 0.001d^2$$

where

$h$ is the height of the golf ball above the ground in metres, and

$d$ is the horizontal distance of the golf ball from point $A$ in metres.

The graph of this equation is drawn below.




**(i)** What is the maximum height the ball reaches above the ground? ***(1 mark)***

**(ii)** There are two occasions when the golf ball is at a height of $35 \text{ metres}$.
What horizontal distance does the ball travel in the period between these two occasions? ***(1 mark)***

**(iii)** What is the height of the ball above the ground when it still has to travel a horizontal distance of $50 \text{ metres}$ to hit the ground at point $B$? ***(1 mark)***

**(iv)** Only part of the graph applies to this model.

Find all values of $d$ that are not suitable to use with this model, and explain why these values are not suitable. ***(2 marks)***

# 14. Algebra, 2UG 2013 HSC 30a

Wind turbines, such as those shown, are used to generate power.

Acknowledgement: Stock Photo–Wind Turbines – Image ID: 1051412 © Miguel Saavedra

1.

In theory, the power that could be generated by a wind turbine is modelled using the equation

$$T = 20\ 000w^3$$

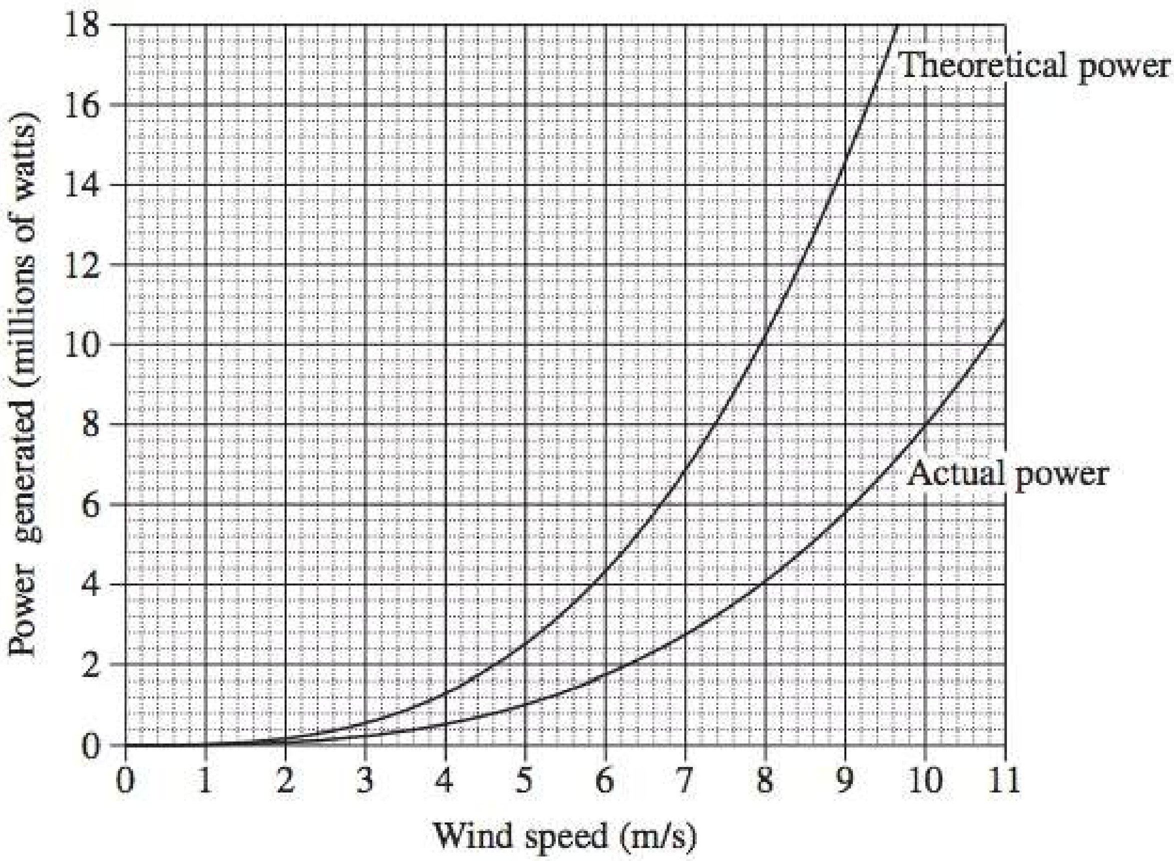
where $T$ is the theoretical power generated, in watts

$w$ is the speed of the wind, in metres per second.

**(i)** Using this equation, what is the theoretical power generated by a wind turbine if the wind speed is $7.3\ \mathrm{m/s}$ ? ***(1 mark)***

**(ii)** In practice, the actual power generated by a wind turbine is only $40\%$ of the theoretical power.
If $A$ is the actual power generated, in watts, write an equation for $A$ in terms of $w$. ***(1 mark)***

The graph shows both the theoretical power generated and the actual power generated by a particular wind turbine.




**(iii)** Using the graph, or otherwise, find the difference between the theoretical power and the actual power generated when the wind speed is $9\ \text{m/s}$. ***(1 mark)***

**(iv)** A particular farm requires at least $4.4$ million watts of actual power in order to be self-sufficient.
What is the minimum wind speed required for the farm to be self-sufficient? ***(1 mark)***

**(v)** A more accurate formula to calculate the power $(P)$ generated by a wind turbine is

$$P = 0.61 \times \pi \times r^2 \times w^3$$

where $r$ is the length of each blade, in metres

$w$ is the speed of the wind, in metres per second.

Each blade of a particular wind turbine has a length of $43$ metres.
The turbine operates at a wind speed of $8\ \text{m/s}$.

Using the formula above, if the wind speed increased by $10\%$, what would be the percentage increase in the power generated by this wind turbine? ***(3 marks)***

## 15. Data, 2UG 2008 HSC 28a

The following graph indicates $z$-scores of 'height-for-age' for girls aged $5 - 19$ years.


**(i)** What is the $z$-score for a six year old girl of height $120$ cm? ***(1 mark)***

**(ii)** Rachel is $10$ ½ years of age.

**(1)** If $2.5\%$ of girls of the same age are taller than Rachel, how tall is she? ***(1 mark)***

**(2)** Rachel does not grow any taller. At age $15$ ½, what percentage of girls of the same age will be taller than Rachel? ***(2 marks)***

**(iii)** What is the average height of an $18$ year old girl? ***(1 mark)***

**(iv)** For adults ($18$ years and older), the Body Mass Index is given by

$$B = \frac{m}{h^2} \text{ where } m = \text{mass} \text{ in kilograms and } h = \text{height} \text{ in metres.}$$

The medically accepted healthy range for $B$ is $21 \leq B \leq 25$.

What is the minimum weight for an $18$ year old girl of average height to be considered healthy? ***(2 marks)***

**(v)** The average height, $C$, in centimetres, of a girl between the ages of $6$ years and $11$ years can be represented by a line with equation

$$C = 6A + 79$$

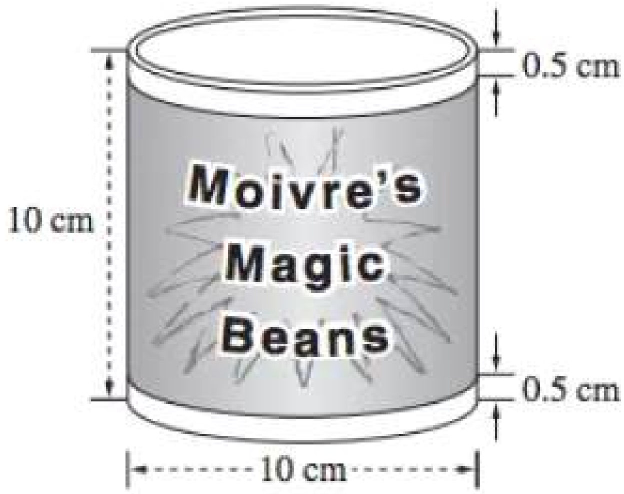
where $A$ is the age in years.

**(1)** For this line, the gradient is $6$. What does this indicate about the heights of girls aged $6$ to $11$? ***(1 mark)***

**(2)** Give ONE reason why this equation is not suitable for predicting heights of girls older than $12$. ***(1 mark)***

## 16. Measurement, 2UG 2010 HSC 28b

Moivre's manufacturing company produces cans of Magic Beans. The can has a diameter of $10 \text{ cm}$ and a height of $10 \text{ cm}$.



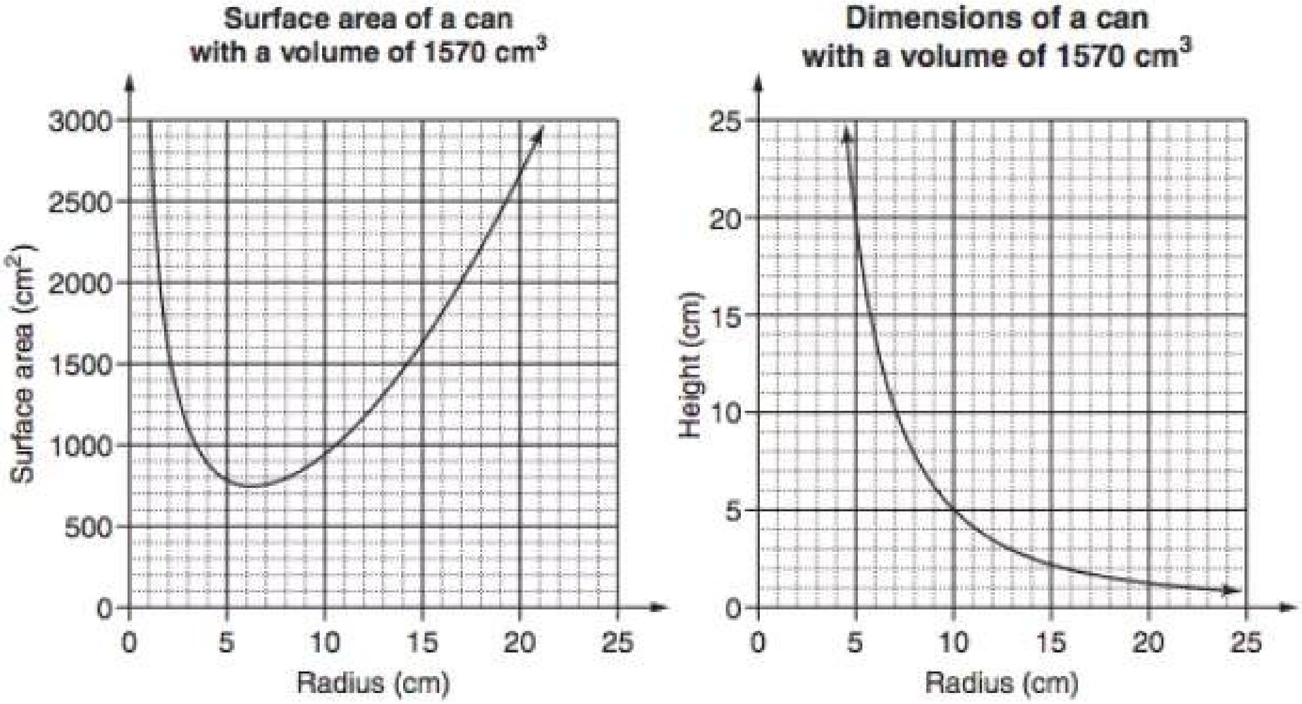

**(i)** Cans are packed in boxes that are rectangular prisms with dimensions $30 \text{ cm} \times 40 \text{ cm} \times 60 \text{ cm}$.
What is the maximum number of cans that can be packed into one of these boxes? ***(1 mark)***

**(ii)** The shaded label on the can shown wraps all the way around the can with no overlap.
What area of paper is needed to make the labels for all the cans in this box when the box is full? ***(2 marks)***

**(iii)** The company is considering producing larger cans. Monica says if you double the diameter of the can this will double the volume. Is Monica correct? Justify your answer with suitable calculations. ***(2 marks)***

**(iv)** The company wants to produce a can with a volume of $1570 \text{ cm}^3$, using the least amount of metal.

Monica is given the job of determining the dimensions of the can to be produced. She considers the following graphs.




What radius and height should Monica recommend that the company use to minimise the amount of metal required to produce these cans? Justify your choice of dimensions with reference to the graphs and/or suitable calculations. ***(2 marks)***

# Worked Solutions

## 1. Algebra, 2UG 2008 HSC 4 MC

$y = 3^x$ passes through $(0, 1)$

and is exponential

$\Rightarrow\ D$

## 2. Algebra, 2UG 2014 HSC 3 MC

Graph is a parabola that

passes through $(0, 1)$

$\Rightarrow\ C$

## 3. Algebra, 2UG 2011 HSC 6 MC

♦ *Mean mark 48%*

At $x = 0,\ y = 1$

As $x \uparrow,\ \ y \uparrow$ exponentially

$\Rightarrow D$

## 4. Algebra, 2UG 2009 HSC 16 MC

♦ *Mean mark 38%*

$$T \propto \frac{1}{S}$$

$$T = \frac{k}{S}$$

As S $\uparrow$ , T $\downarrow$ $\Rightarrow$ cannot be B or D

C is incorrect because it graphs a linear relationship

$\Rightarrow\ A$

## 5. Algebra, 2UG 2010 HSC 13 MC

♦ *Mean mark 50%*

$$\text{Time to melt (T)} \propto \frac{1}{\text{Temp}}$$

$$\Rightarrow T = \frac{K}{\text{Temp}}$$

When $T = 8 \text{ hrs}, \quad \text{Temp} = 30$

$$8 = \frac{K}{30}$$

$$K = 240$$

When $\text{Temp} = 12$

$$T = \frac{240}{12}$$

$$= 20 \text{ hours}$$

$\Rightarrow B$

## 6. Algebra, 2UG 2013 HSC 22 MC

♦♦♦ *Mean mark 24% (lowest mean of any MC question in 2013 exam)*

$\text{Length of garden} = (20 - 2x)$

$$\text{Area} = x(20 - 2x)$$

$$= 20x - 2x^2$$

$\Rightarrow D$

## 7. Algebra, 2UG 2008 HSC 19 MC

$$\text{Height } (h) \propto \sqrt{\text{Termites } (T)}$$

$$\Rightarrow h = k\sqrt{T}$$

Given $h = 35$ when $T = 2000$

$$35 = k \times \sqrt{2000}$$

$$k = \frac{35}{\sqrt{2000}}$$

$$= 0.7826...$$

Find $h$ when $T = 10\,000$

$$h = 0.7826... \times \sqrt{10\,000}$$

$$= 78.26... \text{ cm}$$

$$\Rightarrow B$$

## 8. Algebra, 2UG 2009 HSC 28c

♦♦ *Mean mark 22%*
**CRITICAL STEP:** Reading the first line of the question carefully and establishing the relationship $h = kd^2$ is the key part of solving this question.

$$h \propto d^2$$

$$h = kd^2$$

When $h = 1.6$, $d = 4.5$

$$\therefore 1.6 = k \times 4.5^2$$

$$k = \frac{1.6}{4.5^2}$$

$$= 0.07901\ ...$$

Need to find $h$ when $d = 15$

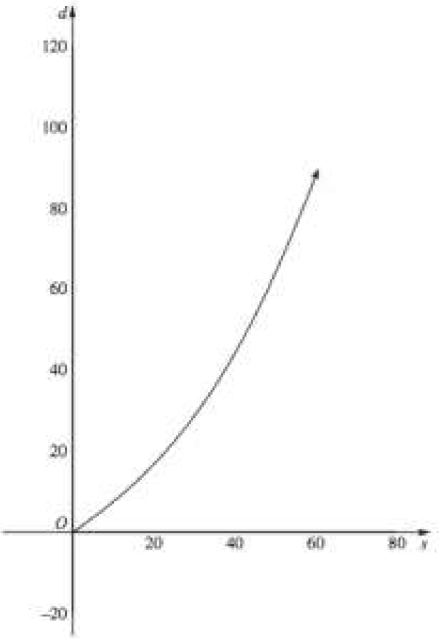
$$h = 0.07901... \times 15^2$$

$$= 17.777...$$

$$= 17.8 \text{ m} \quad \text{(to 1 d.p.)}$$

## 9. Algebra, 2UG 2009 HSC 28a

**(i)**




**(ii)** When $s = 40$

$$d = 0.01(40)^2 + 0.7(40)$$
$$= 16 + 28$$
$$= 44 \text{ m}$$

When $s = 70$

$$d = 0.01\ (70)^2 + 0.7(70)$$
$$= 49 + 49$$
$$= 98 \text{ m}$$

$$\therefore \text{Difference} = 98 - 44$$
$$= 54 \text{ metres}$$

♦ *Mean mark 41%*
**COMMENT:** Students could easily have used the graph for calculating $d$ at $40\ \text{km/h}$, although the formula was required when the speed increased to $70\ \text{km/h}$.

## 10. Algebra, 2UG 2011 HSC 28a

**(i)** $$P \propto \frac{1}{V}$$
$$= \frac{a}{V}$$

♦ *Mean mark 39%*
**COMMENT:** Students ***must*** be able to express a proportional relationship in terms such as $P \propto \frac{1}{V}$ and convert this to an equation $P = \frac{k}{V}$.

**(ii)** When $P = 3, V = 2$

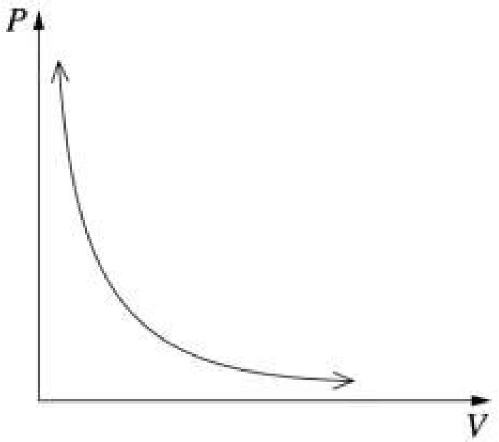
$$\Rightarrow 3 = \frac{a}{2}$$
$$a = 6$$

Need to find $P$ when $V = 4$

$$P = \frac{6}{4}$$
$$= 1\frac{1}{2}$$

♦ Mean mark *47%*

**(iii)**




♦♦ *Mean mark 26%*
**COMMENT:** An inverse relationship is reflected by a hyperbola on the graph.

## 11. Algebra, 2UG 2012 HSC 30c

**(i)** 2030 occurs at $n = 20$ in $x$-axis

Expected population (no policy) $= 6\ 600\ 000$

**(ii)** $A$ represents the population at $n = 0$

which is the population in 2010.

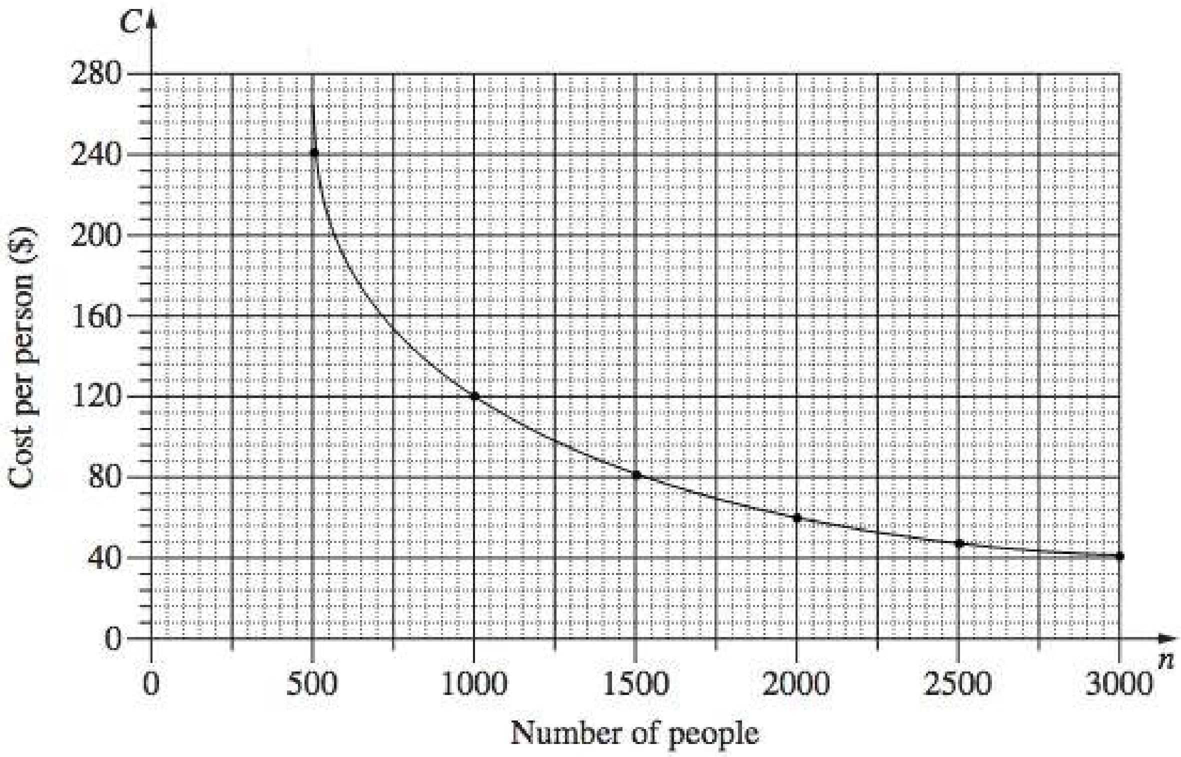
♦ *Mean mark 48%*

## 12. Algebra, 2UG 2014 HSC 29a

**(i)**

| Number of people ($n$) | 500 | 1000 | 1500 | 2000 | 2500 | 3000 |
|---|---|---|---|---|---|---|
| Cost per person ($C$) | 240 | 120 | 80 | 60 | 48 | 40 |

**(ii)**




**(iii)** $C = \dfrac{120\ 000}{n}$

♦ *Mean mark 48%*

**(iv)** Limitations can include:

- $n$ must be a whole number
- $C > 0$

♦♦♦ Mean mark 7%
**COMMENT:** When asked for limitations of an *equation*, look carefully at potential restrictions with respect to *both the domain and range*.

**(v)** If $C = 94$

$$\Rightarrow 94 = \frac{120000}{n}$$

$$94n = 120000$$

$$n = \frac{120000}{94}$$
$$= 1276.595...$$

$\therefore$ Cost cannot be \$94 per person, because $n$ isn't a whole number.

♦ *Mean mark 38%*

## 13. Algebra, 2UG 2012 HSC 30b

**(i)** Max height $= 40$m

**(ii)** From graph

$h = 35$ when $x = 30$ and $x = 170$

$\therefore$ Horizontal distance $= 170 - 30$
$= 140$m

**(ii)** Ball hits ground at $x = 300$

$\Rightarrow$ Need to find $y$ when $x = 250$

From graph, $y = 17.5$m when $x = 250$

$\therefore$ Height of ball is 17.5m at a horizontal distance of 50m before $B$.

**MARKER'S COMMENT:** Responses in the range $17 \leq h \leq 18$ were deemed acceptable estimates read off the graph.

**(iv)** Values of $d$ not suitable.

If $d < 0$, it assumes the ball is hit away from point $B$. This is not the case in our example.

If $d > 300$, $h$ becomes negative which is not possible given the ball cannot go below ground level.

♦♦♦ *Mean mark 12%*
**MARKER'S COMMENT:** Many students did not refer to the domain $d > 300$ as unsuitable to the model. Be careful to examine the *whole* domain in similar questions.

## 14. Algebra, 2UG 2013 HSC 30a

**(i)** $T = 20\,000w^3$

If $w = 7.3$

$$\begin{aligned} T &= 20\,000 \times (7.3)^3 \\ &= 7\,780\,340 \text{ watts} \end{aligned}$$

**(ii)** We know $A = 40\,\% \times T$

$$\begin{aligned} \Rightarrow A &= 0.4 \times 20\,000 \times w^3 \\ &= 8000w^3 \end{aligned}$$

♦ *Mean mark 34%*

**(iii)** Solution 1

At $w = 9$

$A = 5.8$ million watts (from graph)

$T = 14.6$ million watts (from graph)

$$\begin{aligned} \text{Difference} &= 14.6 \text{ million } - 5.8 \text{ million} \\ &= 8.8 \text{ million watts} \end{aligned}$$

♦ *Mean mark 38%*

Alternative Solution

At $w = 9$

$$\begin{aligned} T &= 20\,000 \times 9^3 \\ &= 14\,580\,000 \text{ watts} \end{aligned}$$

$$\begin{aligned} A &= 8000 \times 9^3 \\ &= 5\,832\,000 \text{ watts} \end{aligned}$$

$$\begin{aligned} \text{Difference} &= 14\,580\,000 - 5\,832\,000 \\ &= 8\,748\,000 \text{ watts} \end{aligned}$$

**(iv)** Find $w$ if $A = 4.4$ million

$$\begin{aligned} 8000w^3 &= 4\,400\,000 \\ w^3 &= \frac{4\,400\,000}{8000} \\ &= 550 \\ \therefore\ w &= \sqrt[3]{550} \end{aligned}$$

♦♦ *Mean mark 25%*
**COMMENT:** Students need to be comfortable in finding the cube roots of values - a calculation that can be required in a number of topic areas and is regularly examined.

$= 8.1932...$

$= 8.2 \text{ m/s} \quad (1 \text{ d.p.})$

$\therefore$ The minimum wind speed required is 8.2 m/s

**(v)** Find $P$ when $w = 8$ and $r = 43$

$P = 0.61 \times \pi \times r^2 \times w^3$

$= 0.61 \times \pi \times 43^2 \times 8^3$

$= 1\,814\,205.92$ watts

When speed of wind $\uparrow 10\,\%$

$w' = 8 \times 110\,\% = 8.8 \text{ m/s}$

Find $P$ when $w' = 8.8$

$P = 0.61 \times \pi \times 43^2 \times 8.8^3$

$= 2\,414\,708.08$ watts

Increase in Power $= 2\,414\,708.08 - 1\,814\,205.92$

$= 600\,502.16$

$\Rightarrow$ % Power increase $= \dfrac{600\,502.16}{1\,814\,205.92}$

$= 0.331$

$= 33\% \quad$ (nearest %)

♦ *Mean mark 41%*
**MARKER'S COMMENT:** Students are reminded that a % increase requires them to find the difference in power generated at different wind speeds and divide this result by the original power output, as shown in the Worked Solution.

## 15. Data, 2UG 2008 HSC 28a

**(i)** $z$-score $= 1$

**(ii) (1)** If 2 ½ % are taller than Rachel

$\Rightarrow z$-score of $+2$

$\therefore$ She is 155 cm

**(2)** At age 15 ½, 155 cm has a

$z$-score of $-1$

68% between $z = 1$ and $-1$

$\Rightarrow$ 34% between $z = 0$ and $-1$

50% have $z \geq 0$

$\therefore$ % Above $z$-score of $-1$

$= 50 + 34$

$= 84\%$

$\Rightarrow$ 84% of girls would be taller than Rachel at age 15½.

**(iii)** Average height of 18 year old has $z$-score $= 0$

$\therefore$ Av height $= 163$ cm

**(iv)** $B = \dfrac{m}{h^2}$

$h = 163 \text{ cm} = 1.63 \text{ m}$

Given $21 \leq B \leq 25$, minimum healthy weight occurs when $B = 21$

$$\Rightarrow 21 = \frac{m}{1.63^2}$$

$$\begin{aligned} m &= 21 \times 1.63^2 \\ &= 55.794... \\ &= 55.8 \text{ kg (1 d.p.)} \end{aligned}$$

**(v)** **(1)** It indicates that 6-11 year old girls grow, on average, 6cm per year

**(2)** Girls eventually stop growing, and the equation doesn't factor this in.

## 16. Measurement, 2UG 2010 HSC 28b

**(i)** $\text{Maximum \# Cans} = 4 \times 3 \times 6$

$= 72$ cans

♦♦ *Mean mark 27%*

**(ii)** $\text{Label Area (1 can)} = 2\pi rh$

$= 2 \times \pi \times 5 \times 9$

$= 90\pi$

$= 282.7433...$ cm²

♦ *Mean mark 38%*
**MARKER'S COMMENT:** Many students didn't account for the clearance of 0.5 cm at the top and bottom of each can.

$\therefore$ Label Area (72 cans) $= 72 \times 282.7433...$

$= 20\ 357.52...$

$= 20\ 358$ cm² (nearest cm²)

**(iii)** Original volume $= \pi r^2 h$

$= \pi \times 5^2 \times 10$

$= 785.398...$ cm³

If the diameter doubles, radius $= 10$cm

New volume $= \pi \times 10^2 \times 10$

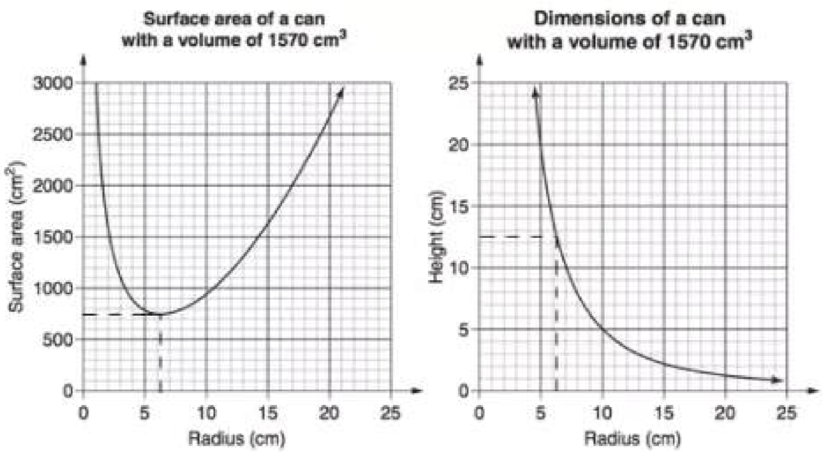
$= 3141.592...$ cm³

$\therefore$ Monica is incorrect because the volume doesn't double. It increases by a factor of 4.

♦ *Mean mark 44%*
**MARKER'S COMMENT:** Many students performed calculations in this part without concluding if Monica is correct or not. *Read the question carefully.*

**(iv)**

♦♦ *Mean mark 26%*




Minimum metal used when surface area is a minimum.

From graph, minimum surface area when $r = 6.3$ cm

When $r = 6.3$ cm, $h = 12.6$ cm (from graph)

$\therefore$ She should recommend radius 6.3 cm and height 12.6 cm